LA HW

Included exercises: 3, 5, 8, 9, 15, 34, 42, 44, 83–87

Exercises 1–16: Matrix–Vector Products

Compute each indicated matrix–vector product.

Original Exercise 3 Compute

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & -1 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}.$$

Original Exercise 5 Compute

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}.$$

Original Exercise 8 Compute

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Original Exercise 9 Compute

$$\begin{bmatrix} s & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & u \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Original Exercise 15 Compute

$$\left(\begin{bmatrix} 3 & 0 \\ -2 & 4 \end{bmatrix}^T + \begin{bmatrix} 1 & 2 \\ 3 & -3 \end{bmatrix}^T \right) \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Exercises 29–44: Linear Combinations

For each exercise, a vector $\mathbf{u}$ and a set S are given. If possible, write $\mathbf{u}$ as a linear combination of the vectors in S.

Original Exercise 34

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}.$$

Original Exercise 42

$$\mathbf{u} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}, \qquad S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Original Exercise 44

$$\mathbf{u} = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \qquad S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} \right\}.$$

Exercises 83–87: Properties of Matrix–Vector Products

Supply the requested proofs. The referenced statements are from Theorem 1.3.

Original Exercise 83 Prove Theorem 1.3(b): for an $m \times n$ matrix A, a vector $\mathbf{u} \in \mathbb{R}^n$, and a scalar c,

$$A(c\mathbf{u}) = c(A\mathbf{u}) = (cA)\mathbf{u}.$$

Original Exercise 84 Prove Theorem 1.3(c): if A and B are $m \times n$ matrices and $\mathbf{u} \in \mathbb{R}^n$, then

$$(A + B)\mathbf{u} = A\mathbf{u} + B\mathbf{u}.$$

Original Exercise 85 Prove Theorem 1.3(d): if $A = [\mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n]$, then

$$A\mathbf{e}_j = \mathbf{a}_j, \qquad j = 1, 2, \ldots, n.$$

Original Exercise 86 Prove Theorem 1.3(e): if B is an $m \times n$ matrix such that $B\mathbf{w} = A\mathbf{w}$ for every $\mathbf{w} \in \mathbb{R}^n$, then $B = A$.

Original Exercise 87 Prove Theorem 1.3(f): for every $m \times n$ matrix A, $A\mathbf{0}$ is the zero vector in $\mathbb{R}^m$.